PROJECT REPORT

Course- Fundamentals of AI and ML

Course Code- CSA 2001

Project Name- ScreenSense AI

"Analyses screen habits using AI"

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Introduction-

In today's digital world, screens have become a regular part of our daily lives. Whether it's smartphones, laptops, or other devices, most people end up spending **more than 6 hours a day** in front of screens—often without even noticing it. While technology makes life easier in many ways, too much screen time can slowly turn into a habit that affects both mental and physical well-being. This growing issue is often referred to as *digital addiction*, where people find it difficult to control their device usage even when it starts causing negative effects.

ScreenSense AI is a digital wellness application designed to help users become more aware of their screen time habits and improve them. It uses a machine learning model to predict addiction risk levels, along with a scoring system that gives users a clear idea of their usage patterns. Based on this, the application provides simple, personalized insights and practical suggestions to help users build healthier digital habits.

This project brings together basic AI techniques and an understanding of user behavior to create a tool that not only tracks screen time but also encourages users to make small, meaningful changes in their daily digital life.

Problem Statement-

The Digital Addiction Crisis

Digital addiction is becoming a common issue today, affecting people of all age groups. As screen usage increases, it starts impacting different areas of life, often without people realizing it.

Some of the major effects of excessive screen time are:

Area	Impact
Physical Health	Poor sleep, eye strain, less physical activity, headaches

Area	Impact
Mental Health	Increased stress, anxiety, low mood, and reduced attention span
Academic/Professional	Lower productivity, procrastination, and difficulty focusing
Social Life	Reduced real-life interaction and less time with family
Personal Well-being	Less time for hobbies, exercise, and self-reflection

My Solution

To overcome these limitations, I developed **ScreenSense AI**.

The goal of this project is not just to track screen time, but to help users understand their habits and improve them. The application combines:

- Machine learning to predict addiction risk levels
- Behavioural insights to provide meaningful and practical suggestions
- A simple and user-friendly interface so that anyone can use it easily

Objective-

The main goal of this project was to build a system that can help users understand and improve their digital habits. To achieve this, I focused on developing a machine learning model capable of predicting digital addiction risk levels (Low, Medium, and High). Along with this, I designed a scoring system on a scale of 0 to 100 to give users a clear idea of their addiction level. The project also aims to generate personalized insights based on individual usage patterns and provide practical suggestions that users can follow to improve their habits. Another important objective was to include a progress tracking feature so users can monitor their improvement over time and stay motivated.

In addition to these primary goals, I also worked on some supporting features to enhance the overall experience. This includes creating simple and interactive visualizations to help users better understand their daily habits, implementing local data storage to ensure user privacy, and preparing proper documentation along with screenshots to present the project in a clear and professional manner.

Approach-

1. Overall Development Process

I followed a step-by-step and iterative approach while building this project. Instead of completing everything at once, I divided the work into phases—starting from understanding the problem, then moving to data collection, model development, UI design, testing, and finally improvements and documentation. At each stage, I tested the results and made changes wherever needed, which helped me gradually refine the project.

2. Data Collection Strategy

Since real user data was not available due to privacy concerns, I created a synthetic dataset with more than 20 examples. While designing the dataset, I made sure it represented realistic usage patterns across different risk levels.

- Low risk users: less screen time, good sleep, and higher study hours
- Medium risk users: moderate screen usage with balanced habits
- High risk users: high screen time, low sleep, and lower productivity

This approach helped in training the model to recognize different behavior patterns effectively.

3. Model Selection

I chose the Decision Tree Classifier for this project because it is simple, easy to understand, and works well with small datasets. One of the main advantages is that it clearly shows how decisions are made, which makes it easier to explain the model. It also does not require complex preprocessing like feature scaling and allows quick training, which was useful during testing and improvement.

4. Scoring Formula Design

Along with prediction, I developed a scoring system (0–100) to represent the user's overall digital habits. The idea was to balance both negative and positive factors.

- Screen time, social media, and gaming increase the score
- Sleep, study, exercise, and family time decrease the score

Different weights were assigned based on their real-life impact. For example, social media was given higher importance as it is more addictive, while study and sleep were treated as strong positive factors. A base value was added to keep the score within a proper range.

5. Risk Level Thresholds

To make the results easier to understand, I divided the score into three categories:

- Low risk → healthy habits
- Medium risk → needs improvement
- High risk → requires immediate attention

These thresholds were adjusted after testing different cases to ensure balanced and realistic results.

6. UI Development Approach

For the user interface, I used Streamlit because it allowed me to build the application quickly using Python. It provides simple and interactive components like sliders, buttons, and charts, which made it easy to collect input and display results. Another advantage is that it does not require separate frontend development, making the whole process faster and more efficient.

Dataset Description

Dataset Overview

The dataset used in this project is stored in the file *data/dataset.csv* and contains more than 20 labelled examples representing different digital usage patterns.

Since real user data could not be used due to privacy concerns, I created this dataset manually while keeping real-life behavior in mind.

Each data entry includes information about daily screen time, time spent on social media, sleep duration, and study or work hours. These features were chosen because they directly reflect a person's digital habits and overall lifestyle. Based on these inputs, each record is assigned a label indicating the level of digital addiction—Low, Medium, or High.

Sample Data Description

For example, a user with low screen time, good sleep, and high study hours is categorized as low risk. On the other hand, a user with very high screen usage, low sleep, and minimal productive time is categorized as high risk. Moderate patterns fall under the medium risk category. This way, the dataset covers a balanced range of usage behaviours.

Data Characteristics

Overall, the dataset is small but carefully designed to represent realistic patterns. It contains around 20 or more examples with 4 main input features and one target label. Even though the data is synthetic, it closely follows real-world trends, which makes it suitable for training and testing the model. The dataset is stored in CSV format, making it easy to use with Python libraries like Pandas.

Model & Implementation

1. Machine Learning Model

The core of ScreenSense AI is a Decision Tree model built using Python's scikit-learn library. I trained it on a small dataset with 20+ examples that represent different types of digital habits. The model uses inputs like screen time, social media usage, sleep, and study hours to predict whether a user falls into low, medium, or high addiction risk. I chose this model because it is simple, easy to understand, and works well with small datasets.

2. Scoring System & Risk Levels

Along with prediction, I created a scoring system that gives a value between 0 and 100 based on the user's daily habits. Activities like screen time, social media, and gaming increase the score, while positive habits like sleep, study, exercise, and family time reduce it. Based on this score, users are categorized into low, medium, or high risk, which helps them quickly understand their situation.

3. Insights & Suggestions

To make the results more useful, the application analyzes each input and provides personalized insights. It highlights critical issues, gives warnings where needed, and also shows positive habits. Based on this, it suggests simple and practical steps that users can follow, including a short improvement plan to help them build better habits gradually.

4. Progress Tracking & Visualization

The app also allows users to track their progress over time by saving recent results locally. This helps them see whether their habits are improving or not. In addition, visual charts are used to display all the habits clearly, making it easier to compare different activities and understand patterns.

5. User Interface & Integration

The entire system is connected through a simple Streamlit interface. Users can enter their data using sliders, click a button to analyze, and instantly see their results. Everything—from input to prediction, scoring, insights, and visualization—is integrated into a single, easy-to-use interface.

Technology Stack

Component	Technology	Purpose
Frontend	Streamlit	Interactive web interface

Component	Technology	Purpose
ML Model	scikit-learn	Decision Tree classifier
Data Processing	Pandas	Dataset handling
Visualization	Matplotlib	Bar charts and graphs
Storage	JSON	Local history tracking
Language	Python 3.8+	Core programming

Results & Output

1. Functional Features

The application works smoothly and includes all the key features that were planned. It calculates a score between 0 and 100 based on seven daily habits and uses this to detect the user's risk level (Low, Medium, or High) with clear color coding. It also provides personalized insights, practical suggestions, and a structured 7-day improvement plan. In addition, users can view their past results and track their progress over time. The AI model prediction and visual charts are also fully integrated and working properly.

2. Scoring Behavior

The scoring system produces realistic and meaningful results. For example, users with healthy habits like low screen time, good sleep, and high productivity usually fall into the low-risk category. Moderate usage patterns result in medium risk, while excessive screen time combined with poor sleep and low productivity leads to high risk. This shows that the scoring system reflects real-life behavior quite well.

3. Model Performance

The Decision Tree model performs well for this project, with an approximate accuracy of 85–95% on the dataset. It uses four main input features and classifies users into three categories: Low, Medium, and High. Even though the dataset is small, the model is able to give consistent and reasonable predictions.

4. User Interface & Output Display

The application has a clean and easy-to-use interface. Users can enter their data using interactive sliders and get results instantly. The output includes the addiction score, risk level, lifestyle rating, and total screen time. The insights section highlights critical issues, warnings, and positive habits, while the suggestions section provides clear and actionable steps for improvement.

5. Visualization & Progress Tracking

To make the results more understandable, the app displays a bar chart showing all seven habits, allowing users to easily compare their daily activities. It also includes a history feature that stores recent analyses and shows trends over time, helping users see whether they are improving, staying the same, or getting worse.

Challenges Faced

- **Medium risk not showing**
  Initially, the app only showed low or high risk. I fixed this by adjusting the scoring logic and adding better data for medium cases.

- **Display issues in Streamlit**
  print() and plt.show() didn't work in the app. I replaced them with Streamlit functions, and everything displayed properly.

- **Model import error**
  The app couldn't find the model file at first. I corrected the file path, and the issue was resolved.

- **Images not loading on GitHub**
  Screenshots were not visible due to wrong file names and paths. After fixing them, all images displayed correctly.

- **History not saving**
  The history feature wasn't working due to file issues. I added proper handling, and it started saving data correctly.

- **Auto-update vs button**
  I chose button-based analysis instead of auto-update to make the app simpler and more user-friendly.

Learning Outcomes

Technical Skills Gained

- **Python**
  Started with basic syntax and improved to using libraries like Streamlit, Pandas, and Matplotlib for building a complete application.

- **Machine Learning**
  Earlier, I only had theoretical knowledge. Through this project, I learned how to actually build and use a model (Decision Tree) in a real application.

- **Web Development**
  Had no experience before, but learned to build a full working web app using Streamlit.

- **Data Visualization**
  Moved from basic understanding to creating clear and interactive charts using Matplotlib.

- **Git & GitHub**
  Improved from basic commands to managing a full project with proper commits and version control.

- **Documentation**
  Learned how to write a proper README and project report to clearly explain the project.

What Makes This Project Special

1. **Real-World Impact:** Addresses a genuine problem affecting millions
2. **AI Integration:** Machine learning for prediction and classification
3. **Complete Application:** End-to-end from data to deployment
4. **Professional Documentation:** Comprehensive README, screenshots, and this report
5. **User-Centered Design:** Clean interface, privacy-focused, actionable outputs

Conclusion-

ScreenSense AI shows how machine learning can be used in a simple and practical way to address the growing issue of digital addiction. The project combines a Decision Tree model, a clear scoring system, and an easy-to-use interface to not only predict addiction risk but also give users helpful insights and suggestions to improve their habits.

While working on this project, I gained hands-on experience in building ML models, developing a web application, and solving real-world issues during implementation. It also helped me understand that AI is not just about complex algorithms, but about using technology in a meaningful way to solve everyday problems. Overall, this project has been a valuable learning experience and a strong base for future improvements.